

VANATVAM

Booking Management System

Application Manual

System Guide, Flow Diagrams & API Reference

Vanatvam Booking System — Application Manual

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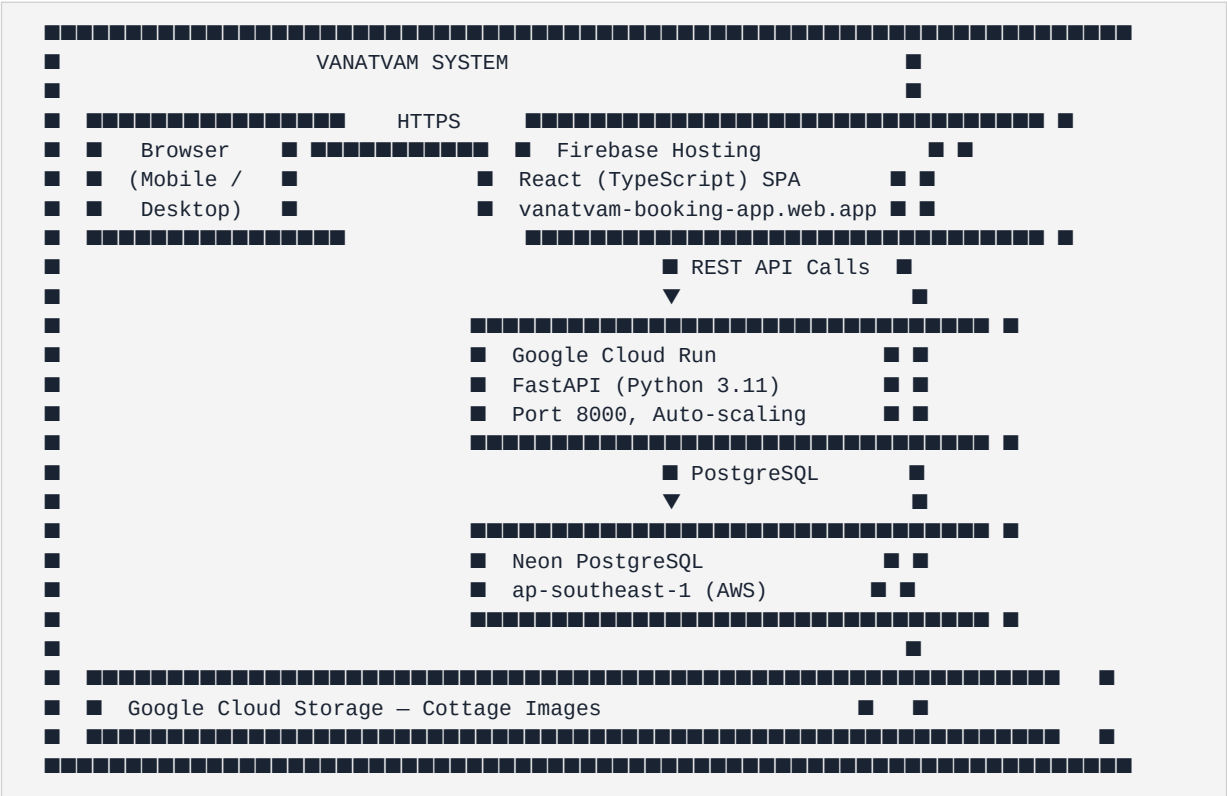
1. System Overview

Vanatvam is a private cottage booking management system for property owners who share access to nature sanctuaries. The system uses a credit-based quota model where each owner receives an annual allocation of weekday and weekend credits to book cottages.

Core Concepts:

- Owners book cottages using credits (not money)
 - Admins manage the entire system — members, properties, bookings, quotas
 - Bookings require admin approval before they are confirmed
 - Holiday and peak season dates consume credits differently (weekend rate)
-

2. Architecture



Tech Stack:

Frontend	React 18, TypeScript, Axios
Backend	FastAPI, Python 3.11, SQLAlchemy
Database	PostgreSQL (Neon serverless)
Auth	JWT (HS256, 30-min expiry)
Hosting	Firebase Hosting + Cloud Run
Storage	Google Cloud Storage (images)
Email	SMTP (configurable via Admin UI)

3. Application Flow Diagram

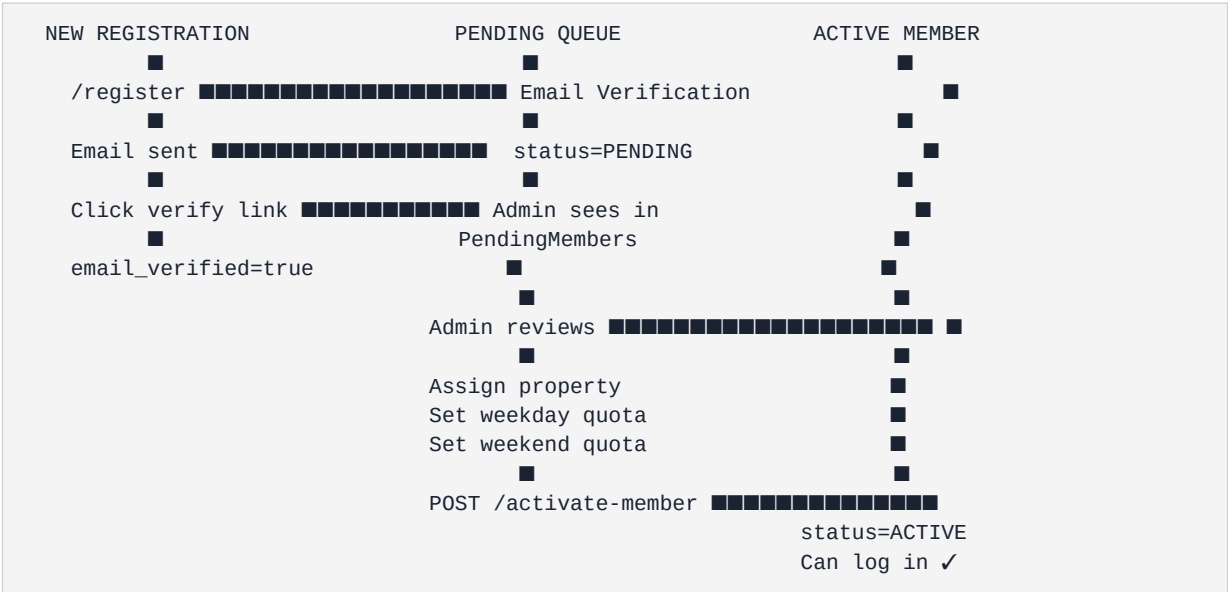
3.1 High-Level System Flow



3.2 Booking Flow



3.3 Member Activation Flow



3.4 Quota System Flow

ANNUAL QUOTA ALLOCATION (Admin sets at member activation)

■ Weekday Balance: e.g. 12 credits/year ■

■ Weekend Balance: e.g. 6 credits/year ■



■ BOOKING MADE ■

■ Each night costs: ■

■ • Mon-Thu (not holiday/peak) → 1 Weekday credit ■

■ • Fri, Sat, Sun → 1 Weekend credit ■

■ • Holiday / Peak Season date → 1 Weekend credit ■



■ BOOKING CANCELLED / REJECTED ■

■ Credits refunded to owner's balance ■



■ ANNUAL RESET (Admin initiates) ■

■ All members reset to their allocated quota ■

■ QuotaReset records transaction history ■

4. User Roles

4.1 Admin

Full system access. Responsible for:

- Approving/rejecting new member registrations
- Managing properties and cottages
- Approving/rejecting booking requests
- Adjusting member quotas
- Configuring holidays and peak seasons
- Blocking maintenance dates
- System reporting and audit trail

4.2 Owner (Member)

Limited access to their own data:

- View available cottages and check availability
- Submit booking requests
- View their booking history
- Check quota balance and transaction history
- Cancel pending/confirmed bookings

5. Authentication Flow

1. REGISTRATION
POST /api/auth/register
Body: { email, password, name, phone }
→ Sends verification email with token
→ User status = PENDING
2. EMAIL VERIFICATION
GET /api/auth/verify-email?token=<token>
→ Sets email_verified = true
→ Admin still needs to activate account
3. LOGIN
POST /api/auth/login
Body: { email, password }
→ Returns: { access_token, token_type: "bearer" }
→ Token stored in localStorage
→ Token expiry: 30 minutes
4. AUTHENTICATED REQUESTS
All API calls include:
Header: Authorization: Bearer <access_token>
5. PASSWORD RESET
POST /api/auth/forgot-password { email }
→ Sends reset link via email
POST /api/auth/reset-password { token, new_password }
→ Updates password

JWT Payload:

```
{
  "sub": "user@email.com",
  "user_id": 123,
  "role": "owner",
  "exp": 1234567890
}
```

6. Owner Workflows

6.1 Making a Booking

1. Navigate to **Dashboard** → **Book a Stay**
2. Select cottage from dropdown (grouped by sanctuary)
3. Set check-in and check-out dates
4. System checks availability (no conflicts, no maintenance)
5. System calculates credit cost breakdown (weekday vs weekend nights)
6. Owner reviews cost and confirms
7. Booking created with status PENDING
8. Credits deducted from balance immediately
9. Admin notified — booking appears in Approval Queue
10. Owner receives email once approved/rejected
11. If rejected, credits are automatically refunded

6.2 Cancelling a Booking

1. Navigate to **My Trips**
2. Find booking with status PENDING or CONFIRMED
3. Click "Cancel booking"
4. Confirm cancellation
5. Status changes to CANCELLED
6. Credits refunded automatically

6.3 Viewing Quota Status

- Navigate to **Quota Status**
 - Shows current weekday balance / total weekday quota
 - Shows current weekend balance / total weekend quota
 - Navigate to **Transaction History** for full credit log
-

7. Admin Workflows

7.1 Activating a New Member

1. Navigate to **Pending Members**
2. Review new registration (name, email, phone)
3. Click Activate
4. Assign a property (sanctuary)
5. Set weekday quota (credits/year, default 12)
6. Set weekend quota (credits/year, default 6)
7. Submit → member status changes to ACTIVE
8. Member receives activation email

7.2 Processing Booking Approvals

1. Navigate to **Approval Queue**
2. Review pending bookings (owner, cottage, dates, credits)
3. Click booking to open decision form
4. Choose **Approve** or **Reject**
5. Add optional notes (mandatory for rejection)
6. Submit → booking status updated, email sent to owner
7. If rejected, owner's credits are automatically refunded

7.3 Managing Properties & Cottages

Properties (Sanctuaries):

- Settings → Sanctuaries tab
- Create/edit sanctuary name and description

Cottages:

- Settings → Cottages tab
- Create cottage: assign to sanctuary, set ID (e.g. C-12), capacity, amenities
- Upload image by clicking or dragging onto the image area
- Filter by sanctuary to manage specific cottages

7.4 Maintenance Blocking

1. Navigate to **Maintenance**
2. Select cottage from dropdown (grouped by sanctuary)
3. Set start and end dates
4. Add reason (optional)
5. System shows warning if existing bookings overlap those dates
6. Submit → cottage blocked for selected period
7. If conflicting bookings exist, use "Revoke" button to cancel them with refund

7.5 Configuring Holidays & Peak Seasons

Holidays:

- Settings → Holidays tab
- Add specific dates with holiday names
- Holiday dates charge weekend credit rate

Peak Seasons:

- Settings → Peak Seasons tab
- Define named seasons with start/end dates
- Peak season dates charge weekend credit rate

7.6 Quota Management

Adjusting a Member's Quota:

- Quota Adjustment → search member → enter weekday/weekend change → submit

Annual Reset:

- Settings → Quota Reset tab
- Resets ALL members' balances back to their allocated quota
- All reset transactions are logged

7.7 Reports & Audit Trail

- **Inventory Health:** Occupancy rates, credit usage statistics
- **Reports** (Calendar → Reports tab): System-wide statistics
- **Audit Trail:** Complete log of all system actions (bookings, activations, quota changes, etc.)
- **Rejected Bookings:** View all rejected/revoked bookings with reasons

8. Credit Quota System

Credit Calculation Rules

Monday – Thursday	1 Weekday credit
Friday	1 Weekend credit
Saturday	1 Weekend credit
Sunday	1 Weekend credit
Any Holiday date	1 Weekend credit
Any Peak Season date	1 Weekend credit

Example:

Booking: Thursday (check-in) to Monday (check-out) = 4 nights

- Thursday: 1 weekday credit
- Friday: 1 weekend credit
- Saturday: 1 weekend credit
- Sunday: 1 weekend credit
- **Total: 1 weekday + 3 weekend credits**

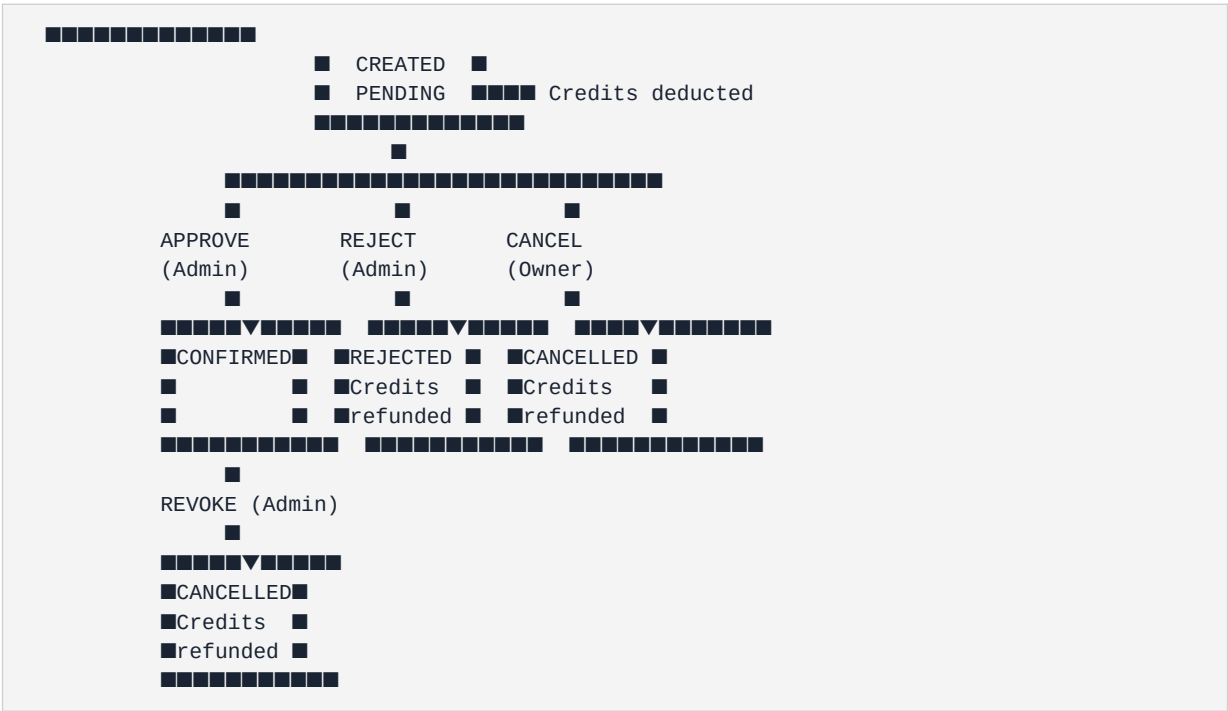
Note: The check-out day is NOT charged or blocked. A guest checking out on Monday does not consume Monday's credit.

Quota Transactions

Every credit movement is logged as a QuotaTransaction:

- activation — initial credits assigned on member activation
- booking — credits deducted when booking submitted
- refund — credits returned on cancellation/rejection
- manual_adjustment — admin manually adjusted balance
- reset — annual quota reset

9. Booking Lifecycle



Status Transitions:

—	PENDING	Owner	Submit booking
PENDING	CONFIRMED	Admin	Approve booking
PENDING	REJECTED	Admin	Reject booking + refund
PENDING	CANCELLED	Owner	Cancel booking + refund
CONFIRMED	CANCELLED	Owner	Cancel confirmed booking
CONFIRMED	CANCELLED	Admin	Revoke booking + refund

10. Calendar & Availability System

Availability Rules (a date is UNAVAILABLE if any of the following):

- 1. **Booked** — an active (pending/confirmed) booking exists for that cottage on that date
- 2. **Maintenance** — a maintenance block covers that date for that cottage
- 3. **Check-out day** — the check-out date of a booking is NOT blocked (new guests can check in)

Calendar Display (Admin Bookings Calendar)

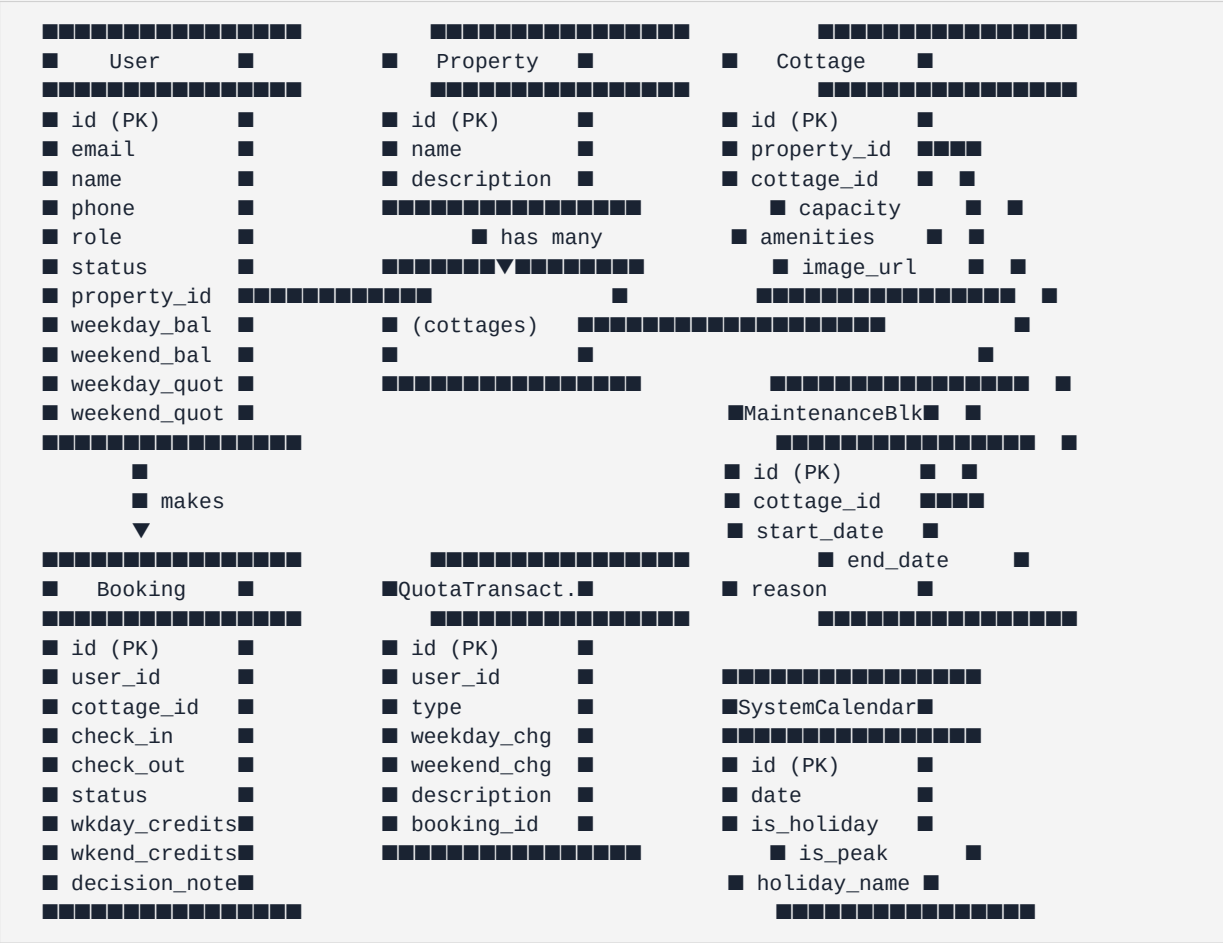
- **Green dot** — confirmed or pending booking on that date
- **Yellow cell** — holiday or peak season date
- **Wrench icon** — maintenance day
- **Today highlighted** in blue
- Click any date to see all bookings for that day

Booking Conflict Detection (Maintenance Blocking)

When creating a maintenance block, the form automatically checks for bookings that overlap the selected dates and displays a warning before submission.

11. Data Models

Entity Relationship Diagram



Key Relationships

User	Property	Many→One
Booking	User	Many→One
Booking	Cottage	Many→One
Cottage	Property	Many→One
MaintenanceBlock	Cottage	Many→One
QuotaTransaction	User	Many→One

QuotaTransaction	Booking	Many→One (opt)

12. API Reference

Base URL

- **Production:** `https://vanatvam-backend-57399834436.asia-south1.run.app`
- **Local Dev:** `http://localhost:8000`

Authentication Endpoints

POST	<code>`/api/auth/register`</code>	None	Register new user
POST	<code>`/api/auth/login`</code>	None	Login, returns JWT token
GET	<code>`/api/auth/me`</code>	JWT	Get current user info
GET	<code>`/api/auth/verify-email`</code>	None	Verify email via token
POST	<code>`/api/auth/forgot-password`</code>	None	Request password reset
POST	<code>`/api/auth/reset-password`</code>	None	Complete password reset

Owner Endpoints

GET	<code>`/api/owner/dashboard`</code>	Dashboard summary
GET	<code>`/api/owner/availability/{cottageid}`</code>	Cottage availability calendar
POST	<code>`/api/owner/calculate-cost`</code>	Preview booking credit cost
POST	<code>`/api/owner/bookings`</code>	Submit booking request
GET	<code>`/api/owner/my-trips`</code>	All owner bookings
POST	<code>`/api/owner/cancel-booking/{id}`</code>	Cancel a booking
GET	<code>`/api/owner/booking-receipt/{id}`</code>	Booking receipt
GET	<code>`/api/owner/quota-status`</code>	Current credit balance
GET	<code>`/api/owner/transactions`</code>	Credit transaction history

Admin Endpoints (selected)

GET	<code>`/api/admin/pending-members`</code>	Members awaiting activation

POST	`/api/admin/activate-member`	Activate member
GET	`/api/admin/approval-queue`	Bookings awaiting approval
POST	`/api/admin/booking-decision`	Approve or reject booking
GET	`/api/admin/check-booking-conflicts`	Pre-check maintenance conflicts
POST	`/api/admin/maintenance-blocks`	Create maintenance block
GET	`/api/admin/bookings-calendar`	Full calendar data
POST	`/api/admin/adjust-quota`	Adjust member credits
POST	`/api/admin/reset-all-quotas`	Annual quota reset
GET	`/api/admin/audit-trail`	System audit log
GET	`/api/admin/reports/statistics`	System statistics

All admin endpoints require JWT token with `role = admin`.

Vanatvam Booking System — Internal Documentation

Last updated: May 2026